Homework 3- Due Date March 09, 11:59 P.M.

Problem 1. Suppose S $= \{2, 3, 4, 5, 6, \cdots\}$ and

$$P(k) = \frac{c}{3^k}, \qquad k = 2, 3, 4, 5, 6, \cdots$$

(a) Find the value of c that makes this a valid probability distribution.
(b) Find P(outcome is even).
(c) Find P(outcome is less than or equal to 6).

Show your work.

Problem 2. When an event A is independent from itself

(a) Always

(b) If and only if $P(A) = 1$

(c) If and only if $P(A) = 0$

(d) If and only if $P(A)$ is either 0 or 1.

Show your work.

Problem 3. Let event M = taking a math class.
Let event S = taking a science class.
Then, M and S = taking a math class and a science class.
Suppose P(M) =0.4, P(S) = 0.6, and P(M AND S) = 0.3.
Are M and S independent?

( ) No
( ) Yes

Show your work.

Problem 4. A player throws darts at a target. On each trial, independently of the other trials, he hits the bull's-eye with probability .06. How many times should he throw so that his probability of hitting the bull's-eye at least once is 0.5?

Problem 5. Two fair six-sided dice are tossed independently. Let X = the maximum of the two tosses.

a. What is the pmf of X?

b. Determine the cdf of X and graph it.

Problem 6. Assume that CDF of a discrete random variable X is given as follows.

$$F(x) = \begin{cases} 0 & x < 1 \\ .30 & 1 \le x < 3 \\ .40 & 3 \le x < 4 \\ .45 & 4 \le x < 6 \\ .60 & 6 \le x < 12 \\ 1 & 12 \le x \end{cases}$$

a. What is the pmf of X?

b. Using just the cdf, compute $P(3 \le X \le 6)$ and $P(4 \le X)$.